

Fields That Use Digital Image Processing

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Introduction to Digital Image Processing

Digital Image Processing

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Image Sources

Categorization of images is often done by energy source:

- Electromagnetic Spectrum (e.g., X-ray, visible light)
- Acoustic/Ultrasonic
- Electron beams
- Synthetic (Computer-generated)

EM Spectrum Overview

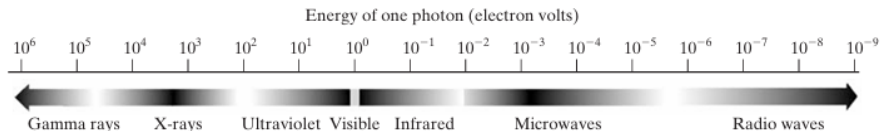


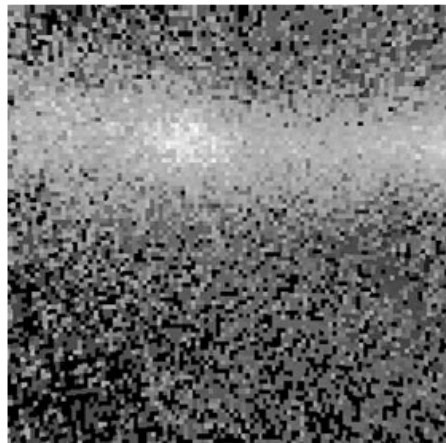
FIGURE 1.5 The electromagnetic spectrum arranged according to energy per photon.

Electromagnetic spectrum arranged by energy per photon

Gamma-Ray Imaging

Applications

- Nuclear Medicine (e.g., PET scans)
- Astronomical Imaging (e.g., Cygnus Loop)
- Reactor Inspection

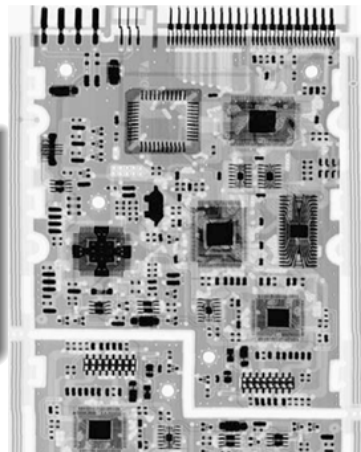


Example: Bone Scan, PET, Cygnus Loop

X-Ray Imaging

Applications

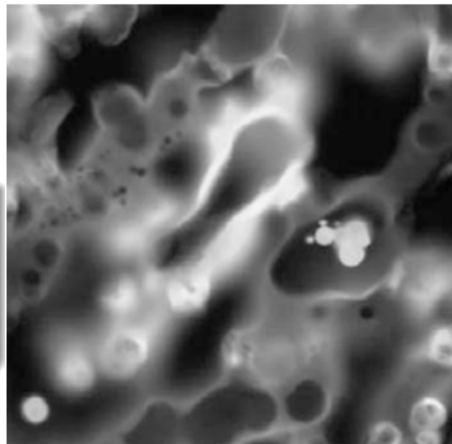
- Medical: Chest X-rays, Angiograms, CT scans
- Industrial: PCB Inspection, Rocket Motor Imaging
- Astronomy



Ultraviolet Imaging

Fluorescence Microscopy

- Highlights fluorescent areas against a dark background
- Used in biology and material sciences

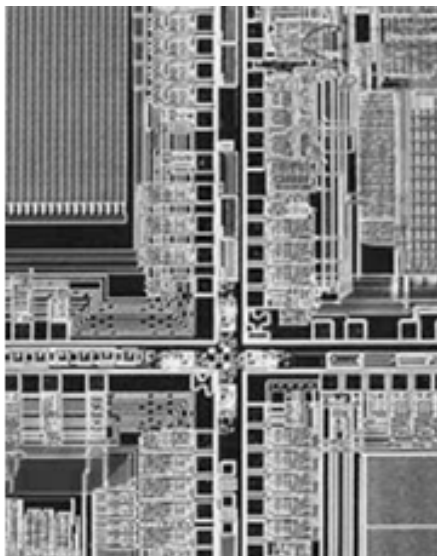


Normal Corn vs. Smut Corn

Visible & Infrared Imaging

Applications

- Microscopy (e.g., pharmaceuticals, microprocessors)
- Remote Sensing (LANDSAT)
- Law Enforcement
- Manufacturing (quality control)



LANDSAT Multispectral Imaging

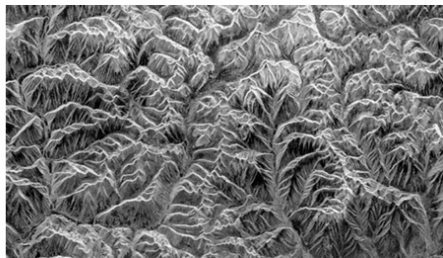
Band	Wavelength (μm)	Uses
1	0.45–0.52	Water penetration
2	0.52–0.60	Plant vigor
3	0.63–0.69	Vegetation detection
4	0.76–0.90	Biomass mapping
5	1.55–1.75	Soil moisture
6	10.4–12.5	Thermal mapping
7	2.08–2.35	Mineral mapping

Thematic Bands in NASA's LANDSAT

Microwave Imaging

Radar Imaging

- Weather-independent
- Can penetrate vegetation and clouds
- Useful in mapping remote regions



Radar Image: Mountains in Tibet

Radio Wave Imaging

Applications

- MRI in Medicine
- Astronomy (e.g., Crab Pulsar)



Conclusion

Key Takeaway

Digital Image Processing touches almost every technical domain — from scanning a human body to analyzing distant stars. Understanding the different energy sources and their applications broadens our grasp of this vast field.

Thank You

Questions?